Book References for Different Topics

Even though multiple book references are shared for different topics, all of the books contain the core same information along with some extra information in different references. Any one book reference (of your choice) would be sufficient to understand the topic.
Important Note: Although all books have same concepts, keep in mind that the different books have different notations for different parameters (e.g. charge is defined by e or q, Electric Field by E or Ɛ etc) and we will follow one of them in the lectures

1. **Crystal Structures**

   Section 1.2.1 (Periodic Structures), 1.2.2 (Cubic Structures), 1.2.3 (Planes and Directions), Section 1.2.4 (The Diamond Lattice) from Streetman book

2. **Two Hydrogen Model**

   Section 1.3.2 from the book "Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson

3. **Silicon Covalent Bonding**

   Section 1.3.3 (Quantum Number and Paulis exclusion principle), and 1.3.4 (Covalent Bonding in Crystalline Solid) from the book "Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson

4. **Finding Carrier concentration**

   Section 3.3 (Carrier Concentrations) from Streetman book, Read Section 2.5 (Intrinsic Semiconductor), 2.6 (Extrinsic Semiconductor) from "Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson

   More details from "Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson

   Section 2.9 (DOS): Mainly to understand background

   Section 2.10: Fermi Distribution

   More on carrier dependence on temperature:

   Section 2.12 (Temperature dependence of carrier concentration)

Section 2.13 (Degenerate Semiconductors)

Section: 2.5.7 (Temperature dependence of carrier concentration) "Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2"

5. **Drift and Diffusion Currents**

   Section 3.2 (Drift Current), 3.3. (Mobility) – for detailed understanding of mobility, 3.4 (Diffusion Current) from "Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson

6. **Constancy of Fermi Level**
   Section 4.1 (Constancy of Fermi Level) from "Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson

7. **Drawing Band Diagram for various materials Refer Class Notes**

8. **Continuity Equation**
   Section - 3.7: Continuity Equations, Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

9. **PN Diode Electrostatics and Ideal IV**
   Using direct diffusion length approximation & Solving continuity Equation
   Section 5.2: Equilibrium Conditions

   Section 5.3: Forward- and Reverse-Biased Junctions; Steady State Conditions
   Book: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee

   Additional Topics:
   6.2 Nonstep Homojunctions (Linearly graded junction)
   6.3 Semiconductor Hetero junctions
      Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

10. **PN Diode Non-ideal IV**
    a) R-G Currents
       Section 5.3.3
       Topic: Generation and Recombination Current in pn Homojunctions (Page 264),
       Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

Section 5.6.2 Recombination and generation in the transition regionBook: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee

For Detailed Derivation:

Section: 6.2.3 The R-G Current, Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2

b) Breakdown

Section 5.3.3

Topic: Reverse Bias Tunneling (Page 270)

Topic: Reverse Bias Carrier Multiplication and Avalanche

Section 5.3.4: Reverse Bias Breakdown

Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

For detailed information

Section: 6.2.2 Reverse Bias Breakdown, Topic: Avalanching, Zener Process

Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2

c) High Level Injection

6.2.4 VA -> Vbi High current phenomena

Topic: High Level Injection

Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2

d) Series Resistance (Ohmic) Effect

6.2.4 VA -> Vbi High current phenomena

Topic: Series Resistance

Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2

11. PN Diode CV (Varactor)

5.5.4 Capacitance of pn-junctions

5.5.5 The varactor diode

Book: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee

More details:

S2.3 Junction Capacitance (page 350) Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

12. Tunnel Diode

Section 10.1 Tunnel Diodes, Book: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee

13. PN Diode Transient Analysis

5.5.1 Time variation of stored charge
5.5.2 Reverse recovery transient
Book: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee
5.5 Transient Effects Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

14. PN Diode: Optoelectronic applications

8.1 Photodiodes, 8.1.2 Solar cells, 8.1.3 Photodetectors
8.2 LIGHT EMITTING DIODES, 8.3 LASERS, 8.4 Semiconductor Lasers Book: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee

Also, PART II: APPLICATION TO OPTOELECTRONIC DEVICES
SOLAR CELL & LED Book: Chenming Hu

15. Narrow Base PN junction diode

8.1.8 The Short diode, Book: Donald Neaman

16. Metal Semiconductor Contacts

5.7 Metal– Semiconductor Junctions Book: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee

4.18 Schottky Diodes: Book: Chenming Hu

Detailed:
6.4 Metal Semiconductor Junctions (Page 323) - Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson
S.2.4 Second Order Effects in Schottky Diodes (Page 348) - Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

17. Bipolar Junction Transistor (BJT)

Chapter 9:Bipolar Junction Transistor (static) 9.1 to 9.4
Book: Fundamentals of Semiconductor Device, Betty Lise Anderson, Richard L. Anderson

18. MOSFET Analysis

Chapter 10: Fundamentals of the MOSFET,
10.1 The two terminal MOS Structure, Book: Donald Neaman
10.2 Capacitance- Voltage Characteristics, Book: Donald Neaman
10.3 The Basic MOSFET Operation, Book: Donald Neaman

Or

Chapter 16: MOS Fundamentals Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2
Chapter 17: MOSFET - The essentials - exclude ac response Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2
Chapter 18: Non-ideal MOS Book: Robert F. Pierret-Semiconductor Device Fundamentals-Addison Wesley (1996)_2

19. Semiconductor Memories

9.5.2 Semiconductor Memories Book: Solid State Electronic Devices (7th Edition), Ben G Streetman, Sanjay Kumar Banerjee